

Pharmacology of Immunosuppressant Drugs, Part 2

Maintenance Immunosuppressive Therapy

- **Calcineurin inhibitors**
- **Antimetabolites**
- **Costimulation blocker**
- **Proliferation signal inhibitors**

Leslie Goldstein, Pharm.D.

Associate Professor

Department of Clinical Specialties

Division of Pharmacology

lgolds01@nyit.edu



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

I am available to groups and individuals for pharmacology help and discussions by appointment.

lgolds01@nyit.edu

On completion of the learning materials and Practice Quiz, the student should be able to:

1. List the pharmacokinetic properties of each class of maintenance drugs for preventing acute allograft rejection and give specific examples.
2. Relate the mechanisms of action of each drug class to its effects on the suppression of cell-mediated and humoral immunity and give specific examples.
3. Describe the mechanisms of toxicities and interactions related to each drug class and give specific examples.
4. Prepare a mind map that integrates the pharmacology of each class and the starred drug with the concepts of immunosuppression therapy.

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Highly relevant optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links

Name of Discipline/ Organ System: Immunology > Treatment of Immune Disorders

Title of Brick: Immunosuppressants

Link: <https://exchange.scholarrx.com/brick/immunosuppressants>

Access Medicine Goodman & Gilman's The Pharmacological Basis of Therapeutics, 15e, 2023;
Chapter 39: Immunosuppressants, Immunomodulation, and Tolerance > Relevant sections

<https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3191§ionid=269169964>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 55:
Immunopharmacology > Relevant sections

<https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/book.aspx?bookid=3382#281742356>

LWW Health Library, Premium Basic Sciences, Lippincott's Illustrated Reviews: Pharmacology
8e, 2023; Chapter 36: Immunosuppressants

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329619&bookid=3222>

Key points

- Maintenance immunosuppression is intended to prevent acute rejection and loss of the allograft while minimizing the risk of infection, malignancy, and drug side effects.
- Maintenance immunosuppressant regimens are combinations of two to four drugs with different mechanisms of action.
- Drugs available for maintenance immunosuppression are calcineurin inhibitors, antimetabolites, glucocorticoids, costimulation blocking agents, and signal proliferation inhibitors.
- The initial maintenance regimen for most patients consists of a calcineurin inhibitor, an antimetabolite, and a glucocorticoid – most frequently tacrolimus, mycophenolate mofetil, and prednisone.
- Modification of the initial regimen may be required due to development of complications, such as toxicity, allograft rejection, infection, malignancy, or pregnancy.
- The drugs have narrow therapeutic windows. Therapeutic drug monitoring for optimizing therapy (optimal levels for efficacy and minimizing toxicity).

Key points

- **The calcineurin inhibitors (CNIs), cyclosporine and tacrolimus**, form a ternary immunophilin-CNI-calcineurin complex within the cytosol, which prevents dephosphorylation and activation of cytosolic NFAT, a transcription factor critical in regulating gene transcription and production of cytokines and T cell receptor-mediated signaling.
 - Calcineurin is a calmodulin-dependent serine/threonine protein phosphatase.
 - Immunophilins are cytosolic isomerases.
 - Cyclosporine binds to cyclophilin. Tacrolimus binds FK binding protein (FKBP).
- CNI side effects include nephrotoxicity, hypertension, hyperlipidemia, hyperglycemia, hyperkalemia, and neurotoxicity.
- CNIs are taken orally once daily (IV formulations are available). They are substrates of CYP3A4 and P-glycoprotein. There is a high potential for drug interactions.
- Tacrolimus is generally preferred. It is associated with lower acute rejection rates and is better tolerated than cyclosporine.

Key points

- **The antimetabolites, mycophenolate mofetil (MMF) and azathioprine (AZA)**, interfere with de novo purine synthesis resulting in inhibition of T cell and B cell proliferation.
- **Mycophenolate** is a potent reversible noncompetitive inhibitor of inosine monophosphate dehydrogenase which blocks the synthesis of guanine monophosphate.
- The most common side effects of MMF include bone marrow suppression, mainly neutropenia, nausea, abdominal cramping, and persistent diarrhea. *Mycophenolate is teratogenic*. MMF is an oral drug (IV available) twice daily. It is hydrolyzed in the liver to active mycophenolic acid, which is excreted in urine as glucuronide metabolites.
- MMF is preferred because of superior efficacy and better side effect profile.
- **AZA** is converted to 6-mercaptopurine and then to the thioinosine monophosphate and 6-thioguanine toxic nucleotides which are incorporated during DNA synthesis.
- The most common AZA side effect is bone marrow suppression manifested by leukopenia. Other side effects, which are seen mainly at higher doses, include nausea, diarrhea, and hepatotoxicity (very high doses). AZA is an oral drug, once daily dosing. 6-MP is inactivated by xanthine oxidase and thiopurine methyltransferase.

Key points

- **Belatacept** is a selective T cell costimulation blocker. It is a recombinant fusion protein of CTLA-4 that binds CD80/86 on antigen presenting cells. It induces immunosuppression by preventing the costimulatory T cell activating signal – binding of CD28 on T cells to CD80/86.
- Belatacept is an alternative to a CNI for prevention of acute allograft rejection in EBV seropositive patients with biopsy-proven CNIs nephrotoxicity. It is given IV every 4 weeks (maintenance phase) in combination with oral MMF and a glucocorticoid.
- Belatacept use is associated with increased risk of EBV-associated post-transplant lymphoproliferative disorder (PTLD); risk is higher in EBV seronegative patients. Common side effects include edema, hypertension, headache, diarrhea, constipation, anemia, and cough.
- Belatacept is **contraindicated** in EBV seronegative patients. Use for liver transplant patients is not recommended due to an increased risk of graft loss and death.
- Information on the use of belatacept in pregnancy is limited.

Key points

- **mTOR inhibitors, sirolimus and everolimus**, are proliferation signal inhibitors. They bind to FK-binding protein and block the activity of mTOR, a cytoplasmic protein kinase, which interrupts DNA and protein synthesis, which prevents progression from the G1 to S phase causing cell cycle arrest, which prevents proliferation of T cells.
- mTOR inhibitors are generally used as alternatives in patients who cannot tolerate CNIs due to toxicity.
- Adverse effects include bone marrow suppression (anemia, leukopenia, thrombocytopenia), dyslipidemia, poor wound healing (antiangiogenic effect), hepatotoxicity, pneumonitis, and increased post-transplant mortality, in addition to nausea, vomiting, and diarrhea or constipation.
- Sirolimus use is not recommended in liver or lung transplantation because safety and efficacy have not been established.
- They are taken orally twice daily. They are substrates of CYP3A4 and P-glycoprotein.
- mTOR inhibitors are contraindicated in pregnancy.

Calcineurin Inhibitors

*Cyclosporine

* Tacrolimus

Calcineurin Inhibitors – Pharmacokinetics

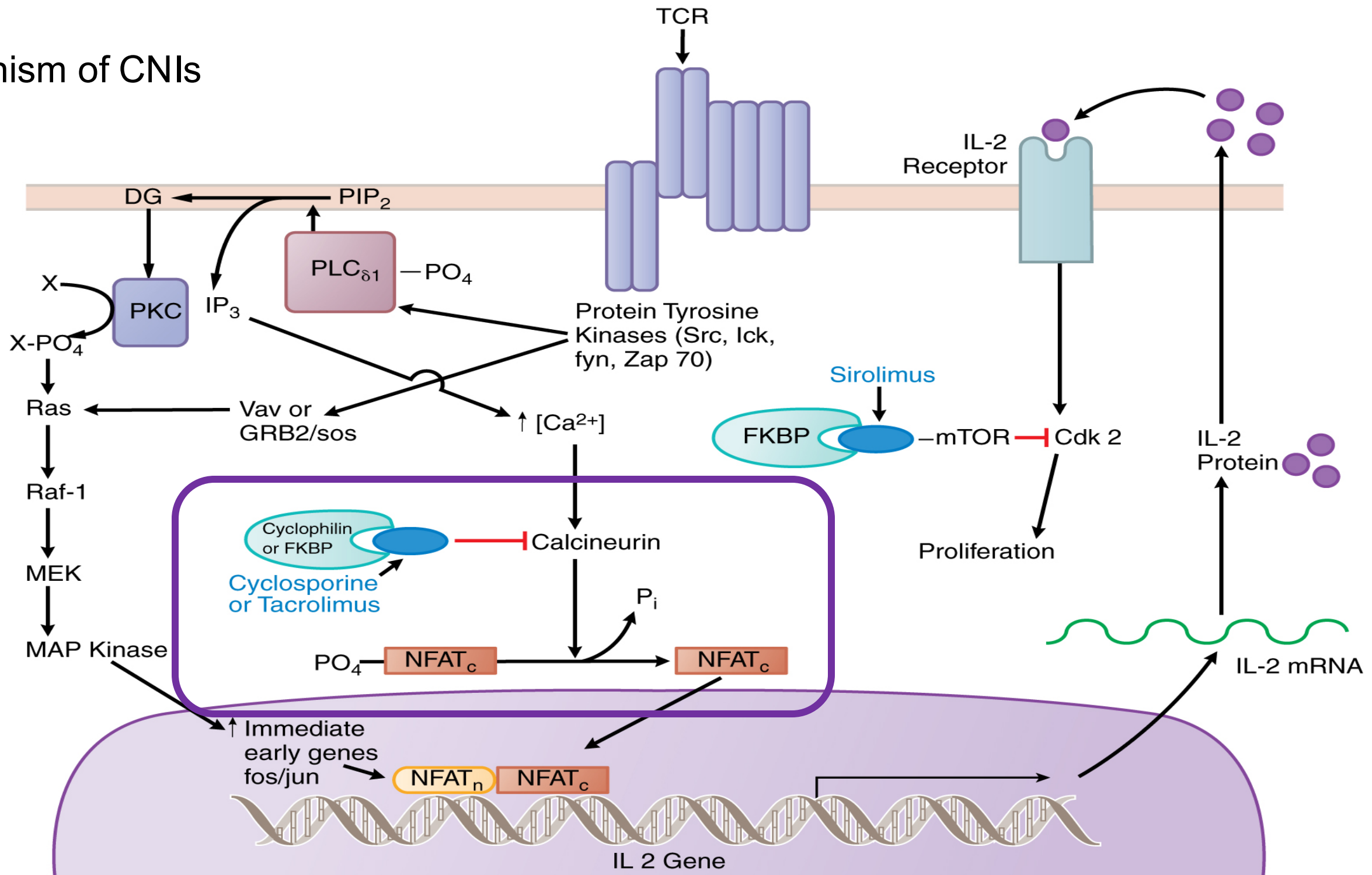
Cyclosporine

Tacrolimus

A	Oral: Incomplete, variable absorption I.V.: solubilized in ethanol-polyoxyethylated castor oil vehicle (and ophthalmic drops)	Oral: Incomplete, variable absorption I.V. (and topical products)
D	90% bound to plasma lipoproteins	99% bound to plasma albumin and α 1-acid glycoprotein
M /E	Substrates of CYP3A4 and P-glycoprotein Biliary / fecal excretion	
t _{1/2}	10-27 hours	3-40 hours

Monitoring plasma concentrations is essential for optimizing therapy.

Mechanism of CNIs



Cyclosporine / Tacrolimus Uses

- Highly efficacious immunosuppressive drugs in routine use
- Prophylaxis of solid organ allograft rejection
- Treatment of GVHD after hematopoietic stem cell transplantation
- Additional uses:
 - Some autoimmune or inflammatory diseases
 - Cyclosporine ophthalmic drops for uveitis
 - Tacrolimus topically for atopic dermatitis, psoriasis

Tacrolimus is preferred by many transplant centers.

Compared with cyclosporine, tacrolimus is associated with decreased acute rejection rates, does not lower mycophenolate levels, is better tolerated by patients, and overall costs are similar.

CNI Toxicities: Cyclosporine / Tacrolimus

Class Toxicities Manifestations

Nephrotoxicity	· Vasoconstrictor effect on afferent renal arteriole · Interstitial fibrosis of renal parenchyma Nephrotoxicity is common in all transplant patients, both renal and nonrenal.	
Hypertension	· Renal vasoconstriction · Sodium retention	
Neurotoxicity	Tremor, headache, insomnia are common. (Other symptoms are less frequent.)	
Hyperkalemia	· Tubular insensitivity to aldosterone · Reduced excretion (K⁺ channel block)	
Hepatotoxicity	· Hyperbilirubinemia · Elevated serum transaminases	
Endocrine	Hyperglycemia (tacrolimus greater risk)	Cyclosporine: · Gingival hyperplasia · Hirsutism
GI	Diarrhea, nausea, vomiting	
Immune system	Infection / malignancy	

Calcineurin Inhibitors: Drug Interactions

High potential

Metabolic and transport related:

- CYP3A4 / P-glycoprotein inducers or inhibitors
- Cyclosporine reduces mycophenolate serum concentrations

Likely mechanism:

Cyclosporine inhibits MRP2 efflux transporter which inhibits biliary excretion of active mycophenolic acid (MPA), thus reduces MPA enterohepatic circulation.

Toxicity related:

Drugs that cause additive toxicity

- Nephrotoxicity
- Neurotoxicity
- Hyperkalemia
- and so on

Note: Cyclosporine and Tacrolimus are not used together – increased risk of toxicities.

CNI Drug Interactions: *High potential*

CYP3A4 and P-glycoprotein Inducers and Inhibitors

Strong Inducers

- Carbamazepine
- Phenobarbital
- Phenytoin
- Rifampin
- St. John's wort
- A few others

Strong Inhibitors

- Ritonavir and other protease inhibitors
- Cobicastat
- Itraconazole and other azole antifungals
- Clarithromycin
- Grapefruit juice
- Many others

Although less potent inducers or inhibitors, many, many drugs can affect the blood levels of CNIs.

- **Practice point:** CNIs have a narrow therapeutic window. Monitoring blood levels is necessary for optimizing therapeutic levels and minimizing toxicity.

Antimetabolites

- * Mycophenolate mofetil (MMF)
- * Azathioprine (AZA)

Mycophenolate mofetil pharmacokinetics and therapeutic uses

Pharmacokinetics:

Mycophenolate mofetil (MMF) is a prodrug of mycophenolic acid (MPA).

- Oral, IV
- MMF is rapidly and completely converted in the liver to active mycophenolic acid (MPA)
- MPA bioavailability 94%
- Enterohepatic circulation of MPA
- Hepatic glucuronidation of MPA → MPA glucuronide (MPAG)
- MPAG excreted in urine
- MPA $t_{1/2}$ mean ~18 hrs

Therapeutic uses:

- Prophylaxis of organ rejection

used in 3-drug combination:

CNI+ MMF + GC

MMF has superior efficacy and better side effect profile than AZA and is the preferred drug for transplantation rejection prophylaxis.

- Prophylaxis and treatment GVHD
- Treatment of psoriasis, lupus, myasthenia gravis, other autoimmune diseases

MPA Sodium enteric coated tablet has different absorption characteristics and may produce less gastric upset.

Mycophenolate mofetil Mechanism of Action

Potent, selective,
uncompetitive,
reversible

IMDPH inhibition

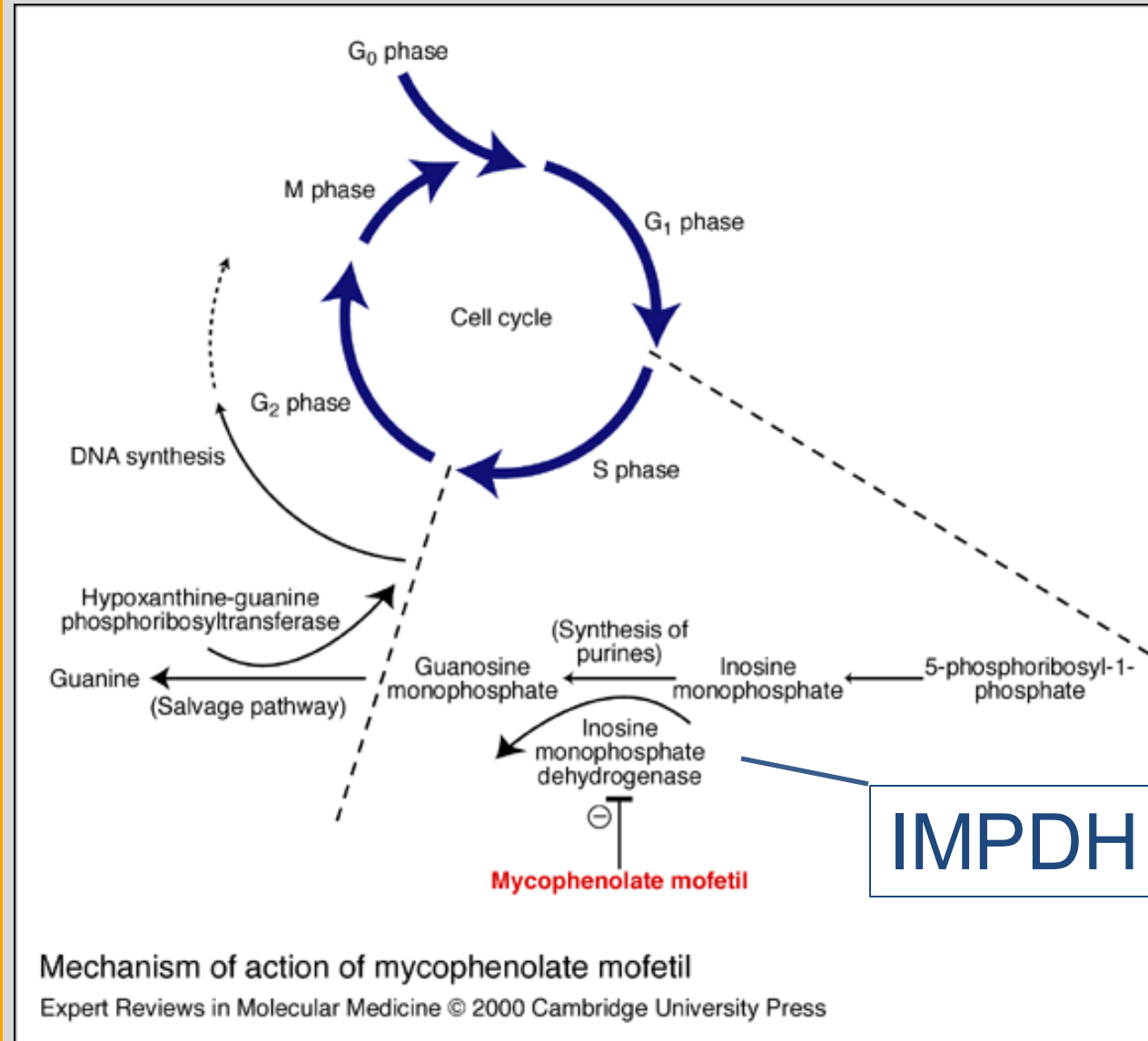
(inosine monophosphate dehydrogenase)



inhibits guanosine nucleotide
synthesis



Suppresses T and B cell
proliferation and function



🔑 T and B lymphocytes are highly dependent on the *de novo* pathway for cell proliferation

Mycophenolate Toxicities

Most common:

- Bone marrow suppression
Anemia and leukopenia most frequent
- Persistent diarrhea, nausea / vomiting
(dose-related)

Risk of serious toxicities is typical of immunosuppressants class including infection and malignancy.

- Pregnancy: Avoid

Increased risk of first trimester pregnancy loss and congenital malformations.

Drug interactions

Avoid or use with caution (monitor).

Immunosuppressants

Enhanced immune suppression increases risk of infection and malignancy

Myelosuppressants

Enhanced bone marrow suppression increases risk of leukopenia, anemia, thrombocytopenia

Cyclosporine

Reduces serum levels of mycophenolic acid.

Cholestyramine Bile acid sequestrants

Chelates drug in gut and reduces absorption

Aluminum / magnesium hydroxide

Antacids reduce absorption. Separate by two hours.

Azathioprine

Pharmacokinetics

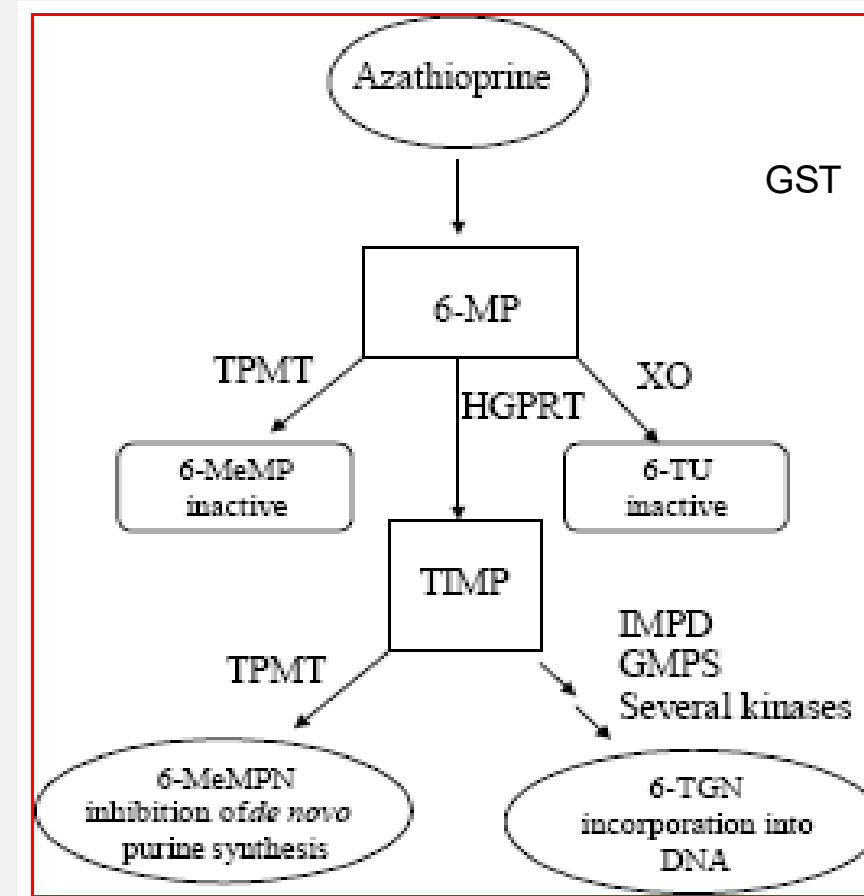
- Oral PRODRUG:
AZA glutathione-S-transferase → 6-MP
- Metabolism to inactive metabolites by
Xanthine oxidase (XO) and
Thiopurine methyltransferase (TPMT)
- Excretion of metabolites in urine
- $t_{1/2}$ of 6-MP ~1 hour

Pharmacogenomics

- Patients deficient in TPMT → at risk for
severe life-threatening myelosuppression

Therapeutic uses

- Prophylaxis of acute allograft rejection
calcineurin inhibitor + AZA + glucocorticoid
- Inflammatory and autoimmune diseases



6-TIMP → 6-TGMP →→ 6-TGTP
↓
6-TGTP is incorporated into DNA of proliferating cells
↓
induces strand breaks and base mispairing
↓
cell death

Azathioprine Toxicities / Drug Interactions

Most serious:

- Bone marrow suppression with profound leukopenia
- Hepatotoxicity with high doses especially in pts with preexisting hepatic dysfunction
- GI effects
- Other potential for toxicities typical of the immunosuppressants class.
- Pregnancy: AZA has been used safely

Allopurinol | Febuxostat
Xanthine oxidase inhibitors

↓ AZA inactivation

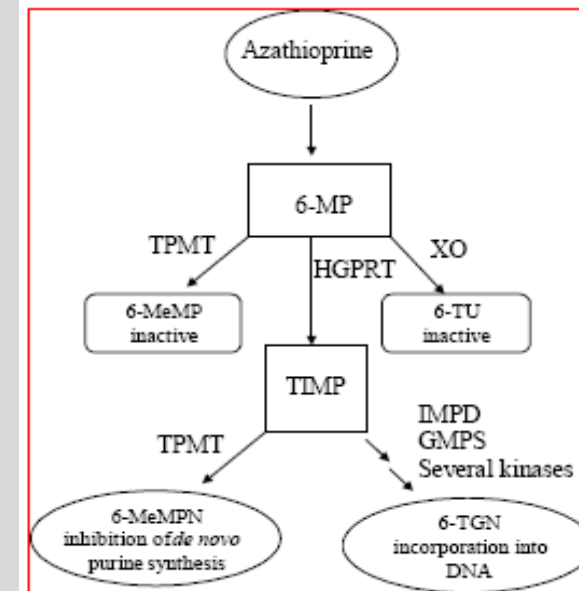


shunt AZA to HGPRT pathway



increase AZA toxicity risk

Remember increased toxicity risk in patients with TPMT deficient activity.



Selective T cell Costimulation Blocker

* Belatacept: 2nd-generation CTLA-4-IgG fusion protein

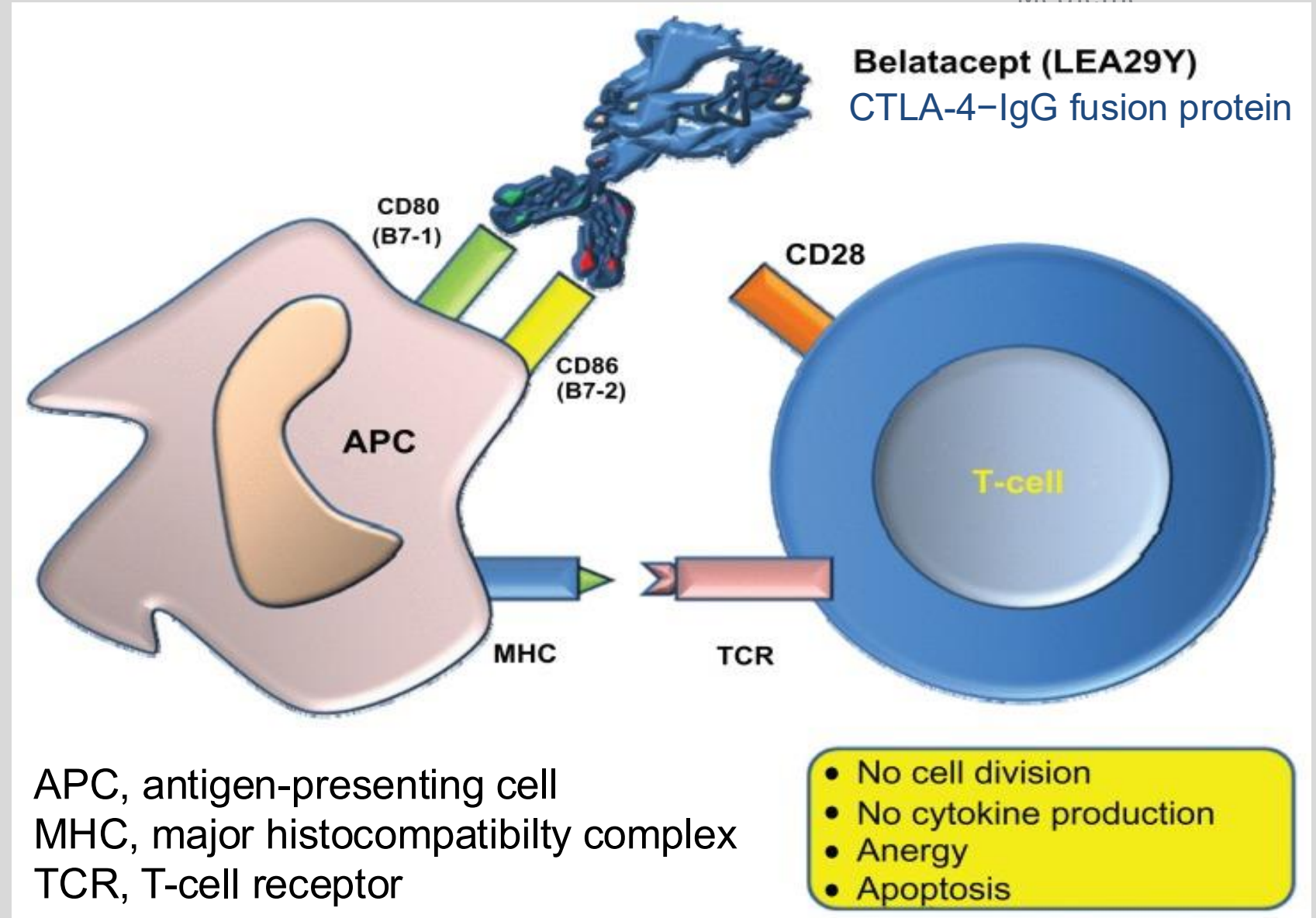
Belatacept

Mechanism of Action

Belatacept binds CD80 (B7-1) and CD86 (B7-2) and blocks costimulation



Inhibits T cell activation, cytokine production, and proliferation



Yale Image Finder: Open Access images and pictures from PubMed Central

Belatacept

Therapeutic uses

- Maintenance therapy for prophylaxis of acute organ rejection in kidney transplant patients who are seropositive for EBV
- Combination with basiliximab, mycophenolate, and corticosteroids

Adverse effects:

- Post-transplant lymphoproliferative disorder (PTLD), primarily involving the CNS
- Reactivation of latent viral infections: Progressive multifocal leukoencephalopathy (PML) by activation of JC virus, and other serious infections
- Other adverse effects typical of the immunosuppressants class and biologic drugs.

Caution / Contraindication:

Belatacept should not be given to EBV-seronegative patient or unknown EBV status. The risk of PTLD is higher for patients who have not been exposed to EBV.

mTOR Inhibitors

*Sirolimus

Everolimus

mTOR: mammalian target of rapamycin
Rapamycin is another name for sirolimus..

Sirolimus Pharmacokinetics

A	Oral (IV for malignant perivascular epithelioid cell tumor) (topical gel for facial fibroangioma associated with tuberous sclerosis)
D	Extensive plasma protein binding; P-glycoprotein substrate Widely distributed in tissues; Partitions into all blood cell types
M	Extensive CYP3A4 metabolism in intestinal wall and liver Bioavailability: tablet ~25%, solution ~14%
E	Fecal excretion (P-glycoprotein efflux in gut) minimal renal excretion
t_{1/2}	average ~60 h

Serum levels should be monitored following initiation of therapy and dosage adjustments.

From: **36 Immunosuppressants**
Lippincott® Illustrated Reviews:
Pharmacology, 7e, 2019, Figure 36.5

Mechanism of action of
mTOR inhibitors, sirolimus and everolimus.

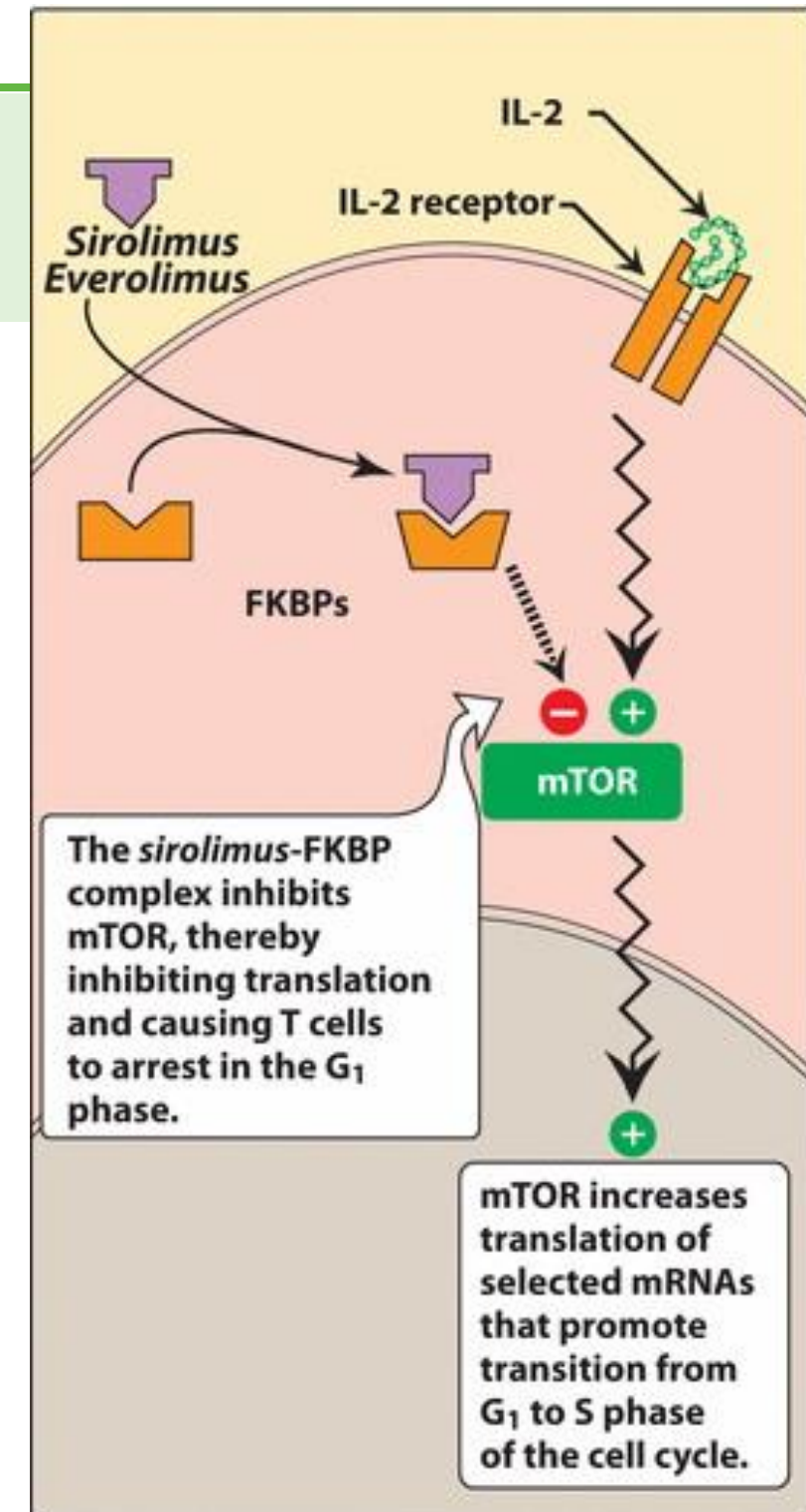
Drug–FKBP–mTOR complex



prevents progression from G₁ to S
phase causing cell-cycle arrest



inhibits cell survival, cell growth, and
proliferation



Sirolimus Therapeutic Uses

Transplant uses:

- Alternative for kidney transplant rejection prophylaxis
- Various (off-label) organ transplantation protocols in patients with CNI-related nephrotoxicity

Safety and efficacy in liver transplantation and lung transplantation have not been established. Use not recommended.

- Acute and chronic graft v. host disease (GVHD) treatment and prevention (off-label)

Non-transplantation uses of sirolimus:

- Drug-eluting coronary stents: Shown to reduce restenosis and additional adverse cardiac events in patients with severe coronary artery disease, due to its antiproliferative effects (prevents smooth muscle proliferation)
- Lymphangioleiomyomatosis: Lung disease caused by abnormal growth of smooth muscle cells in the lungs and lymphatics
- Topical use: Facial angiofibroma associated with tuberous sclerosis

Sirolimus toxicities and Drug Interactions

Adverse effects:

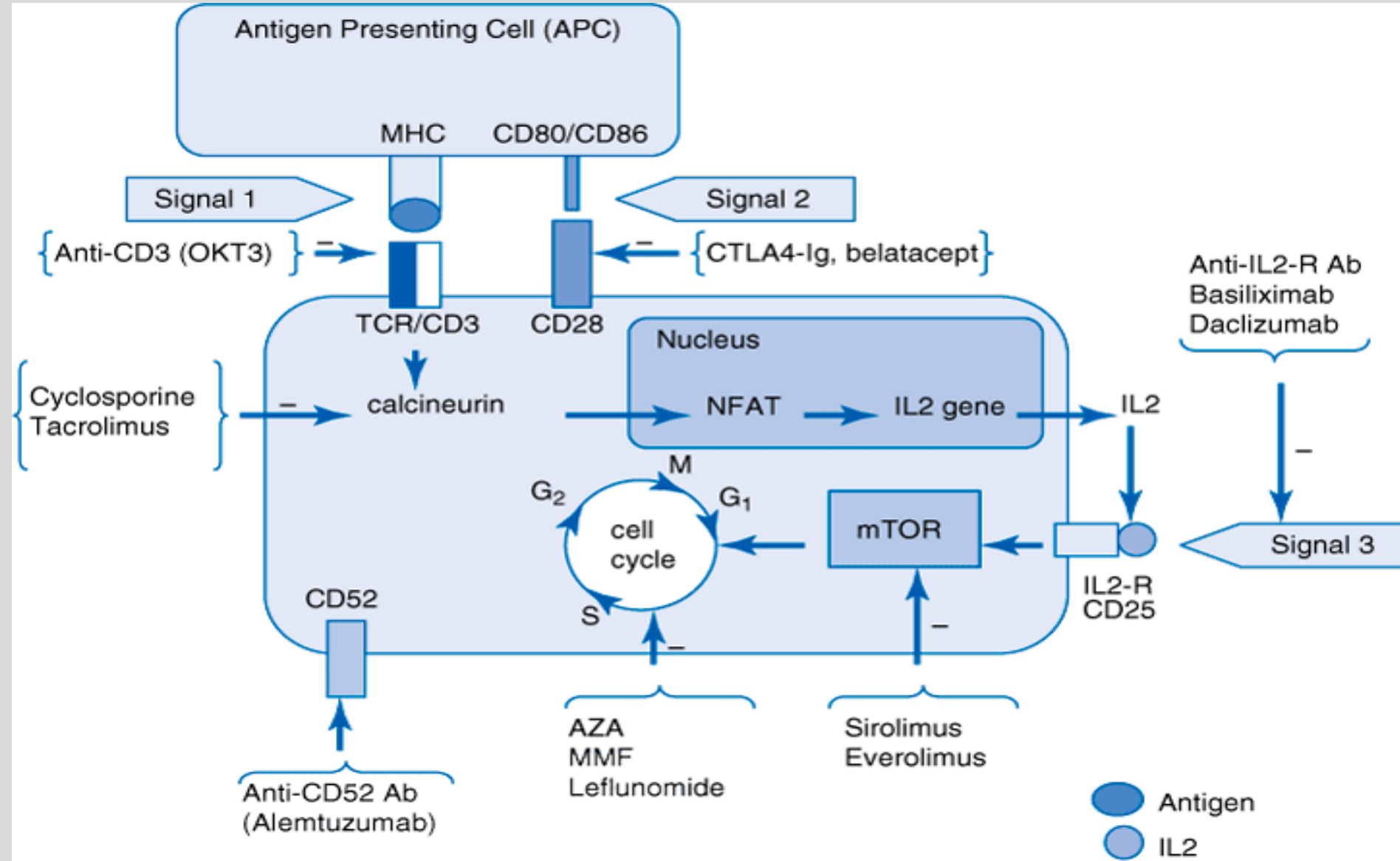
- Profound myelosuppression: Thrombocytopenia (common); anemia; leukopenia
- Impaired wound healing, wound dehiscence
 - May decrease angiogenesis growth factors and fibroblast proliferation.
- Hyperlipidemia: Hypertriglyceridemia and hypercholesterolemia are common.
- Peripheral edema, lymphedema, pleural effusion, pericardial effusions
- Interstitial lung disease (pneumonitis, bronchiolitis obliterans)
- Diarrhea, nausea, vomiting; headache; arthralgia; aphthous ulcers; acne
- Infection risk increased
- Hypersensitivity reactions

Cyclosporine or Tacrolimus + Sirolimus: Avoid combination

- ↑ CNI nephrotoxicity with deterioration of renal function
- ↑ risk of adverse drug events, including mortality
- Cyclosporine may ↑ risk of CNI-induced hemolytic uremic syndrome
- Sirolimus decreases tacrolimus blood levels and AUC (30% decrease)
- CYP3A4 / P-gp inducers / inhibitors

Summary of Immunosuppressants

Transplantation Pharmacology Summary of Therapeutic Targets



Source: Lerma EV, Berns JS, Nissenson AR: *CURRENT Diagnosis & Treatment: Nephrology & Hypertension*: <http://www.accessmedicine.com>
Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

- Maintenance immunosuppressive agents are intended to provide adequate immunosuppression to prevent allograft rejection and minimize infection, malignancy, and drug-related adverse effects. Regimens typically include 2 to 4 drugs.
- The main classes are the calcineurin inhibitors, antiproliferative antimetabolites, costimulation blockers, and proliferation signaling inhibitors (mTOR inhibitors).
- The calcineurin inhibitors are cyclosporine and tacrolimus, which block signal transduction through the calcium-calcineurin pathway, thus impairing T cell activation. Calcineurin is a calcium-dependent protein phosphatase that dephosphorylates NFAT, which enters the T cell nucleus and induces transcription and production of cytokines, including IL-2.
- Cyclosporine binds cyclophilin, while tacrolimus binds FKBP-12. The drug-immunophilin complexes inhibit the activity of calcineurin. Tacrolimus is preferred by many transplant centers due to its decreased rate of allograft rejection.

- Cyclosporine and tacrolimus are substrates of the CYP3A family and P-glycoprotein expressed in the GI tract and liver. There is substantial interindividual variation in oral absorption and metabolism of these drugs. Dosing is based on monitoring drug levels.
- Calcineurin inhibitors are associated with a variety of toxicities. Chief among them is nephrotoxicity. As with all immunosuppressants, infections are possible. Gingival hyperplasia and hirsutism are common side effects of cyclosporine.
- Mycophenolate mofetil (MMF) and azathioprine (AZA) block lymphocyte proliferation by inhibiting nucleic acid synthesis. MMF (active mycophenolic acid) is a potent, reversible, noncompetitive inhibitor of IMPDH, which blocks the de novo formation of guanosine monophosphate, blocking T and B cell proliferation. AZA is a prodrug converted to 6-mercaptopurine and then to cytotoxic 6-thioguanine, which is incorporated into nucleic acid chains, blocking further elongation of the DNA. They are used as adjuncts with calcineurin inhibitors with or without a glucocorticoid. MMF has largely replaced AZA due to improved safety and efficacy profile.

- Belatacept is a recombinant fusion protein of CTLA-4. It is a costimulation blocker by binding CD80/86 on APCs, which inhibits T cell activation. It is intended as a substitute for calcineurin inhibitors to avoid the long-term nephrotoxic, cardiovascular, and metabolic complications seen with cyclosporine and tacrolimus. Belatacept increases the risk of PTLD involving the nervous system. Therefore, it is contraindicated in patients who are seronegative to EBV. Serological titers to EBV are typically obtained to confirm exposure.
- The mTOR inhibitors, sirolimus and everolimus, bind FKBP-12. The complex blocks mTOR leading to inhibition of the IL-2 signal transduction pathway blocking it in the G1-S phase, preventing cell cycle progression, which inhibits T cell proliferation. They are frequently used in multidrug regimens to allow reduction of the calcineurin inhibitors. mTOR inhibitors are associated with impaired wound healing/wound dehiscence, leukopenia, thrombocytopenia, anemia, hypercholesterolemia, hypertriglyceridemia, diarrhea, peripheral edema, and other adverse effects. They are substrates of CYP3A4 and P-glycoprotein. They have low bioavailability, extensive hepatic metabolism, and high potential for drug interactions.

Target	Effect	Drugs
Genes	Inhibit gene expression of pro-inflammatory mediators	Glucocorticoids Antimetabolites Alkylating agents
Lymphocytes	Deplete lymphocyte populations and other specific immune cells	Polyclonal antibodies Anti-CD3; Anti-CD20; Anti-CD25; Anti-CD52 Alemtuzumab
Lymphocyte signaling	Inhibit lymphocyte signaling to block activation and expansion of lymphocytes	Cyclosporine Tacrolimus Sirolimus, Everolimus
Cytokine signaling	Neutralize cytokines and cytokine receptors essential for mediating the immune response	Anti-IL-2R α Anti-TNF α agents JAK inhibitors
Costimulation	Block costimulation to induce anergy	Abatacept Belatacept
Cell adhesion	Block cell adhesion to prevent migration and homing of inflammatory cells	Natalizumab Ustekinumab Vedolizumab
Innate immunity	Inhibit innate immunity, including complement activation	Eculizumab



<https://deathcafe.com/953/>

Organ Donation Art Posted by d-sencier@hotmail.co.uk on March 17, 2018, 5:18 p.m.

References

- Access Medicine Goodman and Gilman's The Pharmacological Basis of Therapeutics, 14e, 2023; Chapter 39: Immunosuppressants, Tolerogens, and Immunostimulants
- Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 55: Immunopharmacology
- Lexicomp monographs

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>